

$$2 \quad \mathbf{A} \quad R = s \left(\frac{Et^2}{\rho} \right)^{0.2}$$

$$\text{units of } s = \text{units of } R \left(\frac{Et^2}{\rho} \right)^{-0.2}$$

$$= \text{m} \left(\frac{(\cancel{\text{kg}} \text{ m}^2 \cancel{\text{s}^{-2}}) (\cancel{\text{s}^2})}{\cancel{\text{kg}} \text{ m}^{-3}} \right)^{-0.2}$$

$$= \text{m} (\text{m}^5)^{-0.2} = 1 \text{ (dimensionless)}$$

3. **C** constant acceleration a_0